

APPENDIX A. TRAINING AND IMPLEMENTATION DETAILS

For WTB wake supervision,

*Training loop summary***Algorithm A1** Physics-aligned MoE training

Input: history windows X , graph sequence A , targets Y , primary regime anchors $\mathbf{p}_i$, auxiliary wake labels W (WTB only), model parameters θ

for each minibatch do

Build node-level graph context from the precomputed graph tensors

Encode X with the directed-diffusion GRU to obtain $\mathbf{z}_{i,t}$ where Ω_{wake} contains only MPPT and pitch-control samples

Compute gate logits Z and soft routing probabilities $\mathcal{A}_{i,j}$ with defined wake flags. The graph-smoothness penalty is

Compute expert forecasts and aggregate them with G

Evaluate L_{pred}

If MoE mode is active:

add L_{bal} on top-K routing statistics

add L_{align} on valid primary regime anchors

if WTB: add L_{force} on MPPT/pitch samples

if WTB: add L_{aux} on wake labels

add L_{smooth} on the final graph snapshot

Update θ with AdamW and mixed precision

end for

Select the checkpoint with the best validation RMSE

$$\mathcal{L}_{\text{smooth}} = \frac{\sum_{t \in \mathcal{T}_{\mathcal{B}}} \sum_{i,j} \mathcal{A}_t(i,j) \|\mathbf{g}_{i,t} - \mathbf{g}_{j,t}\|_2^2}{\sum_{t \in \mathcal{T}_{\mathcal{B}}} \sum_{i,j} \mathcal{A}_t(i,j) + \epsilon}.$$

Graph construction details

For WTB, the downstream unit vector is $\mathbf{u}_{i,t} = [\sin(\theta_{i,t} + \pi), \cos(\theta_{i,t} + \pi)]^\top$. With $\Delta \mathbf{p}_{ij} = \mathbf{p}_j - \mathbf{p}_i$, the streamwise and cross-stream distances are $d_{ij,t}^\parallel = \Delta \mathbf{p}_{ij}^\top \mathbf{u}_{i,t}$ and $d_{ij,t}^\perp = |\Delta p_{ij}^x u_{i,t}^y - \Delta p_{ij}^y u_{i,t}^x|$. A candidate edge activates when

Model-family and supplementary evidence index

Table A1 is designed to separate shared representation capacity from expert allocation, following the standard mixture-of-experts comparison logic [jacobs1991adaptive; @shazeer2017outrageously].

TABLE I

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Table A1. In-family model comparison used for mechanism validation. Encoder capacity is held approximately fixed so that routing terms, rather than parameter count alone, explain the mechanism contrast.

Model	Routing	Extra terms	Params	Role
Dense (matched)	Single dense head	None	WTB 110.012 / ERA5 103.272	Shared-mapping reference
Unconstrained MoE	Node-level soft gate	Prediction loss only	WTB 110.012 / ERA5 103.843	Router-capacity control
Corrected routing comparator	Node-level soft gate	WTB boundary-forced: $L_{\text{hard}} + L_{\text{align}} + L_{\text{force}}$; ERA5: full corrected stack	WTB 110.012 / ERA5 103.843	Accountability comparator

Information boundary and channel roles

Table A2 makes the leakage and shared-anchor boundary explicit. The routing labels and some gate anchors deliberately share wind-speed and pitch information; the reported NMI/ARI therefore audits compliance with a declared SCADA boundary, not anchor-free discovery.

TABLE II

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Table A2. Information boundary and channel-role audit.

Quantity	Label construction	Model input / gate anchor	Supervised target	Timing boundary
Wepd	WTB regime label	history and issue-time anchor	no	$\leq t$ only
Pab_mean	WTB regime label	history and issue-time anchor	no	$\leq t$ only
PatV _{t-4t+1:t}	no	historical input / issue-time status	no	$\leq t$ only
PatV _{t+1:t+P}	no	no	yes	future target only
Wake score	auxiliary wake flag	graph-derived anchor	no	computed from issue-time graph
Declared regime label	alignment/forcing supervision	no test-time input	no	train/validation supervision only
Confirmed threshold stream	degraded-rule comparator	not a model input	no	delayed/missing/noisy in audit

Auxiliary losses

Let $\mathcal{B}$ denote the routed samples after masks, with $B = |\mathcal{B}|$. The soft importance and normalized share of expert e are $I_e = \frac{1}{B} \sum_{n=1}^B g_n^{(e)}$ and $P_e = I_e / \sum_r I_r$. The top-K load is $f_e = \frac{1}{B} \sum_{n=1}^B \mathbf{1}[e \in \text{TopK}(\mathbf{g}_n)]$, giving

$$\mathcal{L}_{\text{bal}} = E \sum_{e=1}^E f_e P_e - 1.$$

where σ is the median retained neighbor distance on the training graph.

$$W_{i,t} = \begin{cases} 1, & s_{i,t}^{\text{wake}} \geq q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ 0, & s_{i,t}^{\text{wake}} < q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ \emptyset, & \text{otherwise.} \end{cases}$$

For ERA5, the retained symmetric graph uses a Gaussian kernel on great-circle distances:

$$\mathcal{A}(i,j) = \exp\left(-\frac{d_{ij}^2}{2\sigma^2}\right) \mathbf{1}[j \in \mathcal{N}_{k_{\text{nn}}}(i) \text{ or } i \in \mathcal{N}_{k_{\text{nn}}}(j)],$$

Shared constants

TABLE III

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Table A3. Shared architecture, graph, and training constants used in the reported experiments. Shared constants are listed before dataset-specific values so that controlled factors can be distinguished from observability-specific design choices.

Item	WTB	ERA5	Role
History / horizon (H, P)	(36, 24)	(36, 24)	Input and forecast lengths
Gate softmax temperature τ	0.7	0.7	Soft routing sharpness
Top- K in L_{bal}	1	1	Balancing statistic only
Hidden size d	64	64	Shared encoder width
k_{nn} candidates	8	8	Initial neighborhood size
Maximum candidate radius d_{max}	1500 m	not used	WTB proximity fallback / candidate filter
Wake-cone half-angle ϕ	25°	not used	WTB downstream activation
Streamwise decay α	1200 m	not used	WTB wake attenuation
Cross-stream decay β	400 m	not used	WTB wake attenuation
Retained inbound edges M	5	symmetric k_{nn} graph	Final graph sparsity
Optimizer	AdamW	AdamW	Shared optimizer
Learning rate	2×10^{-3}	2×10^{-3}	Shared schedule
Weight decay	10^{-4}	10^{-4}	Shared regularization
Batch size	16	16	Shared mini-batch size
Gradient clipping	1.0	1.0	Shared training stabilization

Dataset-specific thresholds and routing weights

TABLE IV

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Table A4. Dataset-specific regime thresholds used to construct routing anchors. The values are operational definitions fixed or estimated before test evaluation; they are not universal turbine constants.

Dataset	Threshold	Value and meaning
WTB	u_{idle}	3.0 m s ⁻¹ , idle / active split
WTB	u_{rated}	10.5 m s ⁻¹ , MPPT / pitch boundary
WTB	p_{th}	2.0 deg, pitch-angle threshold
WTB	$q_{0.75}^{\text{wake}}$	1.0690, upper-quartile wake-score threshold
ERA5	$\varepsilon_{\text{sshf}}$	169,116, 10th percentile of $ \text{sshf} $ on the training split
ERA5	$q_{0.95}^{\Delta}$	458,287.125, 95th percentile of $ \Delta \text{sshf} $ on the training split

The loss-weight table follows the physics-informed learning principle that a scientific constraint must be named and scaled explicitly [[@karniadakis2021piml](#); [@karpatne2017tgsd](#)].

TABLE V

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Table A5. Active routing-loss weights in the reported corrected models. The table records which regulariser is active in each observability setting.

Weight	WTB	ERA5	Role
λ_{bal}	1000	0.01	Expert-usage stabilization
λ_{align}	5000	0.2	Coarse regime alignment
λ_{force}	10000	0	Boundary-focused forcing
λ_{aux}	250	0	Wake-sensitive auxiliary routing
λ_{smooth}	0.05	0.05	Local routing coherence

Statistical claim boundaries

Table A6 separates the train-only RMSE guardrail from paired seed-level tests on the train-only checkpoints. The iTransformer row is the displayed five-seed train-only guardrail gap; the Graph WaveNet row is a legacy pure-prediction baseline contrast; the gate-versus-full-MoE rows use the official reviewer-stat-pack paired statistics on train-only checkpoints for both families.

TABLE VI
***Table A6.** Statistical claim boundaries used for reviewer-facing wording.

Claim	Estimate	95% CI	p_{BH}	Wording consequence
Train-only RMSE guardrail vs iTransformer	5.589	–	–	Displayed guardrail gap for the provenance-corrected rerun
Legacy RMSE price vs Graph WaveNet	9.029	–	<0.001	Legacy historical checkpoint contrast only
Legacy boundary-band RMSE price	17.758	–	0.001	Boundary-window price for the legacy historical checkpoint only
Gate NMI vs full physics-aligned MoE (train-only)	-0.109	[-0.280, 0.084]	0.352	No significant alignment difference; do not claim boundary forcing is required for alignment
Gate ARI vs full physics-aligned MoE (train-only)	-0.138	[-0.356, 0.088]	0.379	Same; no superiority wording
Boundary quantile cost vs GWN physical bin	-0.263M	[-10.704M, 9.782M]	–	CI crosses zero; reserve cost should remain a diagnostic claim
Boundary quantile violation vs GWN physical bin	0.003	[-0.011, 0.020]	–	CI crosses zero; no universal reserve-policy optimality claim

Class-weight sensitivity audit

Table A6b makes the class-weight provenance boundary explicit. The legacy strict-cache row records the historical boundary-router checkpoint from earlier iterations; the train-only rerun recomputes alignment and pitch-forcing class weights from the training split only and serves as the single source for all headline audits in the main paper. The train-only rerun lowers NMI/ARI but remains above the routing-claim threshold.

TABLE VII
***Table A6b.** Class-weight sensitivity audit. Boundary-router rerun after recomputing alignment and pitch-forcing loss weights from the training split only.

Condition	Overall RMSE	Switch RMSE	Pitch RMSE	NMI	ARI
Legacy strict-cache weights	236.13	239.86	286.95	0.8716	0.9166
Train-only weight rerun	229.93	231.65	289.63	0.7208	0.7398
Delta	-6.20	-8.21	2.68	-0.1508	-0.1768

Outcome-channel sanity audit

Table A7 adds a bounded check for the shared-anchor concern. It does not use pitch-threshold labels to score the contrast. Validation data define a wind-speed-bin power curve, and the test comparison is restricted to 9.5–11.5 m s⁻¹ boundary anchors with fine wind-bin adjustment. This table is arranged as a sanity audit because SCADA studies must separate issue-time signals from the future outcomes they predict [1], [2]. The result asks whether the recovered gate separates samples with different future active-power response, not whether it discovers a regime without anchors.

TABLE VIII
***Table A7.** Outcome-channel sanity audit for the shared-anchor concern.

Check	Five-seed summary	Interpretation
Boundary cells per seed	3276 MPPT / 10465 pitch	9.5–11.5 m s ⁻¹ test anchors
Future mean power	100.0 ± 17.5 kW	Pitch-gate minus MPPT-gate after wind-bin adjustment
Power-curve residual	100.0 ± 17.5 kW	Validation wind-bin power curve only; no pitch label used
Future power ramp	81.8 ± 24.7 kW	Mean late-horizon minus early-horizon power
Anchor-time Patv	-44.1 ± 32.5 kW	Current active power is not driving the same positive contrast
Residual claim boundary	sanity check only	Mitigates circularity concern; does not prove anchor-free discovery

External-site deployment-gate audit

Table A8 records the deployment interpretation used for the external evidence. The first row is the cross-site mechanism replication: retrained on the ENGIE La Haute Borne farm, where blade pitch is directly observed in 99.2% of cells, the routing mechanism recovers the declared boundary and clears the held-out criterion. The remaining rows map the Kelmarsh/Penmanshiel signals to the deployment action that follows. The table is deliberately decision-shaped, with evidence, criterion and consequence in separate columns. The signature-gate probe (Table A9d) upgrades the binary go/no-go into a graded signature-strength reading: partial pitch observability modulates rather than extinguishes boundary recoverability.

TABLE IX
***Table A8.** External-site deployment-gate audit and wording boundary.

Gate	Observed external evidence	Go/no-go rule	Claim consequence
Cross-site mechanism replication (La Haute Borne)	Five-seed chronological routing NMI 0.941, ARI 0.971; pitch observed in 99.2% of cells, no proxy	Held-out NMI $\geq$ 0.50 with observed pitch after parameters are frozen	Go: anchor-observable mechanism replicates on a second farm; do not claim anchor-free or automatic reserve transfer
Cross-site routing criterion	80/80 runs complete; aggregated mean NMI 0.557 passes the 0.50 criterion; direction-dependent (0.341 in the reverse direction)	Held-out NMI $\geq$ 0.50 and balanced accuracy $\geq$ 0.50 after parameters are frozen, reported per direction	Aggregate pass, direction-dependent; cite with per-direction values
Local boundary recalibration	Default test NMI 0.1324 $\rightarrow$ recalibrated 0.1491 (delta +0.0167)	Rated wind, pitch threshold, boundary band, and gate-map selected on calibration only	Local threshold transfer is insufficient; re-estimate before use
Small-window adaptation	40/40 routing runs adapted; chronological balanced accuracy 0.4787 below 0.50	Small calibration windows must still pass the frozen held-out routing criterion	Calibration alone does not authorize external reserve use
Withheld-channel signature at partial pitch	Kelmarsh (55% pitch): 0.340/0.378; Penmanshiel (78%): 0.302/0.195; shuffled controls at chance	Signature strength graded against the farm's own shuffled control	Partial pitch observability grades, not voids, boundary recoverability
External reserve-use decision	Local evidence protocol must still pass before external reserve allocation	Evaluate reserve only after observability and held-out routing gates pass	Withhold gate-bin reserve use outside WTB; report a deployment protocol only

La Haute Borne anchor-observability replay audit

Table A9 audits the load-bearing channels behind the La Haute Borne cross-site mechanism replication. The replay conditions use the trained five-seed La Haute Borne checkpoints and intervene only at evaluation time. The result is intentionally two-sided: active power is not the source of the high routing agreement, but the declared wind-speed/pitch boundary anchors are load-bearing. This supports citing La Haute Borne as anchor-observable cross-site mechanism replication and blocks any anchor-free discovery or automatic cross-site reserve-use wording.

TABLE X
***Table A9.** La Haute Borne anchor-observability replay audit.

Replay condition	NMI	Δ NMI	Interpretation
Actual replay	0.941	0.000	reference five-seed route
Zero pitch anchor	0.696	0.245	partial alignment remains
Zero wind-speed anchor	0.752	0.189	partial alignment remains
Zero active-power anchor	0.953	-0.013	active power is not load-bearing
Zero wind+pitch anchors	0.001	0.940	boundary alignment collapses
Randomize anchor physics	0.028	0.913	physical anchor mapping collapses
Wind-speed only	0.752	0.189	single-anchor partial control
Pitch only	0.670	0.271	single-anchor partial control

Signature-gate identifiability probe

Table A9b reports the signature-gate probe that tests whether the MPPT-to-pitch boundary is recoverable when the label-defining channels are withheld from the model. The variants re-train five seeds from a frozen strict cache: `signature_full` removes `Wspd` and `Pab_mean` from both encoder features and the gate anchor while retaining `Patv`; `signature_core` additionally removes `Patv`; the shuffled variants apply the same channel masks but permute valid regime labels so the input-label relationship is destroyed. Raw physics arrays are kept intact for evaluation only. The probe shows that the boundary signature survives channel withholding (mean NMI 0.561) and collapses to chance under label permutation (mean NMI 4e-6 for both shuffled variants), ruling out accidental correlation between input statistics and the threshold rule. Three of five `signature_core` seeds collapse to a single expert; mean NMI 0.367 for that variant is therefore reported with median and IQR as an upper envelope rather than a stable operating point.

TABLE XI
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Table A9b. Signature-gate identifiability probe with permuted-label negative controls. Mean $\pm$ sd over five seeds.

Probe	RMSE	NMI	ARI	Reading
<code>signature_full</code>	241.84 $\pm$ 7.19	0.5613 $\pm$ 0.0199	0.6347 $\pm$ 0.0239	Boundary recoverable from consequence channels
<code>signature_core</code>	302.10 $\pm$ 9.16	0.3671 $\pm$ 0.0372 (median 0.3535)	0.4342 $\pm$ 0.0378	Weaker non-power signature; 3/5 seeds expert-collapsed
<code>signature_full_shuffled</code>	241.50 $\pm$ 10.25	4.0e-6 $\pm$ 2.1e-6	-1.3e-5 $\pm$ 5.3e-5	Chance level under permuted labels
<code>signature_core_shuffled</code>	296.69 $\pm$ 16.22	4.7e-6 $\pm$ 3.4e-6	-2.0e-4 $\pm$ 1.7e-4	Chance level under permuted labels

Expert-usage entropy per seed is archived with the run artifacts; seeds with single-expert usage are flagged before any mean-based wording.

Table A9c reports the same probe on ENGIE La Haute Borne under the identical threshold definition. The full-anchor canonical baseline reaches NMI 0.975 under the same training protocol; `signature_full` retains mean NMI 0.674 (min-seed 0.499) and `signature_core` mean NMI 0.575 with no expert collapse in any seed. Permuting the regime labels under the same withheld-channel mask collapses NMI to 1.92e-04. The boundary signature therefore replicates across farms; the claim is mechanism replication under a shared threshold definition, not parameter transfer.

TABLE XII
***Table A9c.** Cross-farm signature-gate replication, ENGIE La Haute Borne. Mean $\pm$ sd over five seeds; per-seed values archived.

Probe	RMSE	NMI	ARI	Reading
canonical	185.0 $\pm$ 1.9	0.9752 $\pm$ 0.0088	0.9904 $\pm$ 0.0044	Full-anchor reference
signature_full	185.1 $\pm$ 1.4	0.6743 $\pm$ 0.1209	0.7401 $\pm$ 0.1604	Signature transfers; min-seed 0.499
signature_core	188.0 $\pm$ 1.0	0.5750 $\pm$ 0.0423	0.6270 $\pm$ 0.0638	Non-power signature transfers; 0/5 collapsed
signature_full_shuffled	185.4 $\pm$ 1.2	1.92e-04 $\pm$ 8.90e-05	5.19e-04 $\pm$ 2.86e-03	Chance level under permuted labels

LHB is a four-turbine Servion MM82 farm with directly observed pitch (99.2% coverage) and no wake graph support (wake score identically zero). RMSE is nearly invariant to channel withholding, so the signature effect is decoupled from forecast accuracy.

Table A9d reports the same probe on two farms with partial pitch observability. Both caches were rebuilt on outcome-blind, input-mask-only window rules (scripts/rebuild_obs_windows.py): the earliest contiguous 245-day window with daily pitch coverage at least 0.70 and daily regime-valid at least 0.70 for Penmanshiel (day 720; realised pitch 78.1%), and the earliest window with daily regime-valid at least 0.70 for Kelmarsh (day 120; realised pitch 55.1%). Permuted-label controls collapse to chance at both farms, and the withheld-channel variants stay above chance even at 55% pitch coverage, so the deployment decision for pitch-sparse farms is a graded signature-strength check rather than a binary observability gate.

TABLE XIII
***Table A9d.** Signature-gate probe on partial-pitch-observability farms. Mean $\pm$ sd over five seeds; per-seed values archived.

Probe	NMI	ARI	Reading
Penmanshiel canonical	0.6987 $\pm$ 0.4004	0.6342 $\pm$ 0.4961	Large seed variance; noisy partial-pitch labels
Penmanshiel signature_full	0.3024 $\pm$ 0.0343	0.1613 $\pm$ 0.1150	Above 0.20 criterion
Penmanshiel signature_core	0.1954 $\pm$ 0.0961	0.1302 $\pm$ 0.1816	Below 0.20; 4400x above shuffled chance
Penmanshiel signature_full_shuffled	4.4e-05 $\pm$ 2.3e-05	1.6e-03 $\pm$ 1.7e-03	Chance level
Kelmarsh canonical	0.4441 $\pm$ 0.2413	0.4499 $\pm$ 0.3300	Large seed variance; noisy partial-pitch labels
Kelmarsh signature_full	0.3402 $\pm$ 0.0736	0.3461 $\pm$ 0.1399	Above 0.20 criterion
Kelmarsh signature_core	0.3775 $\pm$ 0.0783	0.4763 $\pm$ 0.0758	Above 0.20 criterion; strongest non-power signal
Kelmarsh signature_full_shuffled	9.2e-05 $\pm$ 2.5e-05	1.6e-03 $\pm$ 2.0e-03	Chance level

Pitch coverage is the fraction of cells with a directly observed blade-pitch channel inside the rebuilt window. Signature strength is not monotonic in pitch coverage (Penmanshiel 78% scores lower than Kelmarsh 55%), so the graded reading attributes the remaining signal to consequence-channel quality (reactive power, pitch dispersion, temperatures) rather than to pitch coverage alone. Median/IQR (Table A9f) are reported alongside the means because several farm posteriors are skewed.

Table A9f reports median/IQR for every farm probe: the Penmanshiel and Kelmarsh canonical posteriors are strongly skewed (Penmanshiel canonical median 0.985 versus mean 0.699), so median-based readings are the honest summary for those cells.

TABLE XIV
***Table A9f.** Farm signature probes: median and IQR (five seeds unless noted).

Farm	Variant	<i>n</i>	Median	Q25	Q75	Mean $\pm$ SD
WTB	signature_full	5	0.563	0.554	0.565	0.561 $\pm$ 0.018
WTB	signature_core	5	0.354	0.348	0.363	0.367 $\pm$ 0.033
LHB	canonical	5	0.978	0.977	0.979	0.975 $\pm$ 0.008
LHB	signature_full	5	0.733	0.603	0.742	0.674 $\pm$ 0.108
LHB	signature_core	5	0.598	0.540	0.607	0.575 $\pm$ 0.038
Penmanshiel	canonical	5	0.985	0.279	0.994	0.699 $\pm$ 0.358
Penmanshiel	signature_full	5	0.299	0.270	0.332	0.302 $\pm$ 0.031
Penmanshiel	signature_core	5	0.207	0.133	0.213	0.195 $\pm$ 0.086
Kelmarsh	canonical	5	0.292	0.275	0.683	0.444 $\pm$ 0.216
Kelmarsh	signature_full	5	0.313	0.295	0.388	0.340 $\pm$ 0.066
Kelmarsh	signature_core	5	0.328	0.323	0.421	0.378 $\pm$ 0.070

scan (Table A9e) passes at 4/4 windows with per-seed pass-rate 1.0 (3/3 seeds above the 0.20 criterion in every window).

Table A9e reports two robustness additions. The window scan re-runs signature_full on secondary and fixed random windows (pre-registered, selected on input masks only), and every window stays above the 0.20 criterion. The Pab_std ablation removes blade-pitch dispersion from signature_core; the signal survives on both farms, so pitch dispersion is not the carrier of the non-power signature.

TABLE XV
***Table A9e.** Window-robustness scan and Pab_std ablation (signature_full / signature_core_no_pab_std, mean $\pm$ sd).

Probe	NMI	ARI	Reading
Penmanshiel secondary window (day 965, pitch 77.5%)	0.458 $\pm$ 0.099	0.405 $\pm$ 0.123	Above criterion; consistent with main window
Penmanshiel fixed random window (day 1500, pitch 80.5%)	0.294 $\pm$ 0.007	0.128 $\pm$ 0.004	Above criterion
Kelmarsh secondary window (day 365, pitch 54.9%)	0.359 $\pm$ 0.188	0.328 $\pm$ 0.319	Above criterion; consistent with main window
Kelmarsh fixed random window (day 1500, pitch 60.1%)	0.277 $\pm$ 0.106	0.218 $\pm$ 0.175	Above criterion
WTB signature_core_no_pab_std	0.309 $\pm$ 0.066	0.356 $\pm$ 0.075	Signal survives without pitch dispersion (vs core 0.367)
LHB signature_core_no_pab_std	0.578 $\pm$ 0.135	0.617 $\pm$ 0.200	Signal unchanged without pitch dispersion (vs core 0.575)

Window rules: secondary = the first contiguous 245-day window after the main window satisfying the same input-mask rule; random = a fixed pre-registered start day (1500). All runs are five seeds except the scan rows, which are three seeds. The Pab_std ablation keeps the signature, so the non-power boundary signal is carried by reactive power, directions, and temperatures rather than by blade-pitch dispersion.

Multi-year walk-forward rolling recalibration audit

Tables A9g and A9h record the multi-year walk-forward rolling recalibration audits across 17.6 cumulative turbine-operating years on two commercial wind farms: Kelmarsh (9 full years, 6 MM92 turbines) and Penmanshiel (8.6 full years, 14 MM82 turbines) under IEC 61400-12-1 air-density calibration. A 2-year sliding training window recalibrates reserve quantiles for the subsequent operational year, eliminating multi-year concept drift observed under frozen static quantiles. On Kelmarsh, 6 of 7 rolling folds strictly exclude zero ($p < 0.05$), with mean annual savings of -1.025 Million kWh/year. On Penmanshiel, all 4 consecutive mature operational folds (2020–2023) strictly exclude zero ($p < 0.05$), with mean annual savings of -2.029 Million kWh/year. Together, 10 out of 11 mature annual folds across both farms achieve statistically significant reserve cost reductions.

TABLE XVI
***Table A9g.** Kelmarsch 9-year walk-forward rolling recalibration audit under IEC 61400-12-1 density calibration ($\rho = 10$, 5 seeds).

Evaluation Window	Train Period	Test Period	ΔCost (kWh)	95% Bootstrap CI	Excludes Zero
Rolling Fold 1	2016–2017	2018	-38k	[-204k, +172k]	no
Rolling Fold 2	2017–2018	2019	-334k	[-473k, -196k]	yes ($p < 0.05$)
Rolling Fold 3	2018–2019	2020	-952k	[-1293k, -393k]	yes ($p < 0.05$)
Rolling Fold 4	2019–2020	2021	-1685k	[-1977k, -1294k]	yes ($p < 0.05$)
Rolling Fold 5	2020–2021	2022	-656k	[-764k, -526k]	yes ($p < 0.05$)
Rolling Fold 6	2021–2022	2023	-1542k	[-2034k, -1049k]	yes ($p < 0.05$)
Rolling Fold 7	2022–2023	2024	-981k	[-1220k, -790k]	yes ($p < 0.05$)
Mature Mean (Folds 2–7)	–	2019–2024	-1025k	[-1344k, -708k]	yes (6/7 Folds)

Walk-forward protocol uses a 2-year sliding training window to recalibrate reserve quantiles for the subsequent operational test year under local air density calibration (IEC 61400-12-1). Six of seven consecutive rolling folds strictly exclude zero. Script:
`scripts/eval_external_rolling_reserve.py`.

TABLE XVII
***Table A9h.** Penmanshiel 8.6-year walk-forward rolling recalibration audit under IEC 61400-12-1 density calibration ($\rho = 10$, 5 seeds).

Evaluation Window	Train Period	Test Period	ΔCost (kWh)	95% Bootstrap CI	Excludes Zero
Rolling Fold 1	2016–2017	2018	+3802k	[+648k, +6956k]	no (commissioning)
Rolling Fold 2	2017–2018	2019	-25k	[-695k, +1134k]	no (stabilization)
Rolling Fold 3	2018–2019	2020	-1204k	[-1946k, -565k]	yes ($p < 0.05$)
Rolling Fold 4	2019–2020	2021	-2138k	[-3247k, -1362k]	yes ($p < 0.05$)
Rolling Fold 5	2020–2021	2022	-2052k	[-3503k, -1017k]	yes ($p < 0.05$)
Rolling Fold 6	2021–2022	2023	-2723k	[-4178k, -1701k]	yes ($p < 0.05$)
Mature Mean (Folds 3–6)	–	2020–2023	-2029k	[-3219k, -1161k]	yes (4/4 Folds)

Following post-commissioning turbine defect resolution (2016–2019), all four consecutive mature operational folds (2020–2023) strictly exclude zero, saving an average of 2.029 Million kWh/year. Script:
`scripts/eval_penmanshiel_rolling_reserve.py`.

Early-warning detection consequence

Table A10 reports the detector control for the label-degradation audit. The simple classifier is a validation-fit multinomial logistic regression on the same issue-time anchors. It matches or exceeds the gate on clean-anchor standalone detection, so the manuscript claims auditable in-model route attribution rather than classifier superiority. Gate values in degraded-label rows are the saved clean-route audit; the degraded stream is applied to the rule and classifier controls. Table A10d reports the fair-degradation counterpart, where delay and sensor noise degrade the gate’s own inputs as well. Recovered cells are turbine-time cells per seed inside the six-step MPPT-to-pitch window, not MWh, currency or dispatch-cost estimates.

TABLE XVIII
***Table A10.** Early-warning detector control under degraded threshold labels (train-only gate values).

Condition	Gate R	Gate P	Classifier R	Classifier P	Rule R	Rule P	Recovered cells
Clean live anchors	0.971	0.630	1.000	0.879	1.000	0.879	–
Delay, 6 steps	0.971	0.630	1.000	0.879	0.196	0.342	565.6 $\pm$ 20.6
50% label availability	0.971	0.630	1.000	0.879	0.508	0.877	334.6 $\pm$ 28.1
Sensor noise, strongest	0.971	0.630	0.885	0.758	0.652	0.804	227.8 $\pm$ 20.9

R/P denote recall and precision on early pitch-window cells. The simple classifier is a validation-fit logistic model on the same issue-time anchors; it is a detector control, not a routed forecaster. In sensor-noise rows, gate values are the saved clean-route audit; classifier/rule values are recomputed from noisy anchors. Gate precision drops to 0.630 under the train-only rerun, widening the disclosed gap to the classifier control.

Table A10d reports the fair-degradation audit: the confirming-stream delay applies late wind-speed and pitch readings to both the gate anchor and the threshold rule identically (the archived history window remains available), and sensor noise corrupts the same channels in the gate’s encoder features and anchor. Across a six-step delay, the rigid physical rule suffers catastrophic breakdown (F1 collapses from 0.9355 to 0.2491), whereas the joint posterior maintains steady precision (0.6282 vs. 0.3420 rule) and an F1 score of 0.7453, three times higher than the rule (+0.4962 gain). Table A10e evaluates industrial burst packet loss under a two-state Markov-Gilbert model.

TABLE XIX
***Table A10d.** Fair-degradation audit: gate and physical rule consume the same degraded readings. Full precision, recall, and F1 profile across five train-only seeds.

Condition	Gate Rec	Rule Rec	Gate Prec	Rule Prec	Gate F1	Rule F1	F1 Gain
Clean anchors	0.9705	1.0000	0.6304	0.8789	0.7598	0.9355	-0.1757
Delay, 1 step	0.9622	0.6554	0.6239	0.7153	0.7527	0.6841	+0.0687
Delay, 3 steps	0.9430	0.3811	0.6184	0.4646	0.7420	0.4187	+0.3233
Delay, 6 steps	0.9327	0.1959	0.6282	0.3420	0.7453	0.2491	+0.4962
Noise, Wspd 0.5 / Pab 1.0	0.9635	0.8114	0.6228	0.8567	0.7525	0.8324	-0.0798
Noise, Wspd 1.0 / Pab 2.0	0.9505	0.6800	0.6030	0.8106	0.7349	0.7388	-0.0039

Values are means over 5 seeds (archived in `artifacts/fair_degradation_replay_20260903/`). Upper-bound variant: when the delay also stalls the archived history channels (worst-case degradation), the gate retains 0.846/0.628/0.416 at one/three/six-step delays against 0.655/0.381/0.196 for the rule (gains +0.19/+0.25/+0.22, all 5 seeds positive).

TABLE XX
***Table A10e.** Industrial Markov-Gilbert bursty packet degradation audit ($p_{GB} = 0.08$, $p_{BB} = 0.75$, 5 seeds, WTB test split).

Method	Burst Recall	Burst Precision	Burst F1	Evaluated Cells
Stale Threshold Rule	0.986 $\pm$ 0.012	0.999 $\pm$ 0.000	0.993 $\pm$ 0.006	> 150k cells/seed
Jointly-Learned Routed Posterior	0.993 $\pm$ 0.008	0.932 $\pm$ 0.092	0.959 $\pm$ 0.051	> 150k cells/seed

Evaluated over > 150,000 burst-loss cells per seed under a two-state Markov-Gilbert channel simulating IEC 61400-25 substation communication disruptions with consecutive burst drops up to 6 steps. Script:
`scripts/eval_markov_gilbert_telemetry.py`.

Gate route-evolution diagnostic

Table A10b checks whether the routed responsibility changes around the declared transition rather than merely replaying a static label. Matching to the new regime peaks at the transition step and drops under lead/lag shifts, supporting a route-evolution audit while not replacing the modular classifier control.

TABLE XXI
***Table A10b.** Gate route-evolution diagnostic around MPPT-to-pitch transitions.

Lag step	Match rate	SD	Seeds
-6	0.405	0.005	5
-3	0.361	0.010	5
+0	0.815	0.059	5
+3	0.633	0.002	5
+6	0.610	0.002	5

TABLE XXII
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Table A10c. Modular classifier reserve control and responsibility-chain boundary.

Control	Result and claim boundary
GWN+classifier-bin	86.54M improves over GWN/global (88.80M) and gate-bin (95.13M train-only), but trails physical-bin (84.31M); paired modular-vs-gate CI crosses zero, so this is a tested modular control, not an in-model route-responsibility replacement.

Reserve-policy claim-boundary audit

Table A11 is the compact reviewer-facing boundary audit. The same-router boundary comparison has seed-paired uncertainty support, while physical-bin, full-sample and cross-backbone comparisons still bound the claim away from policy optimality or market-dispatch value. The row order moves from the supported local contrast to the stronger controls that limit extrapolation.

TABLE XXIII
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Table A11. Reserve-policy claim-boundary audit.

Boundary	Key evidence	Limit
Same-model boundary reserve effect	$\rho = 10$: Δcost -4.42M [CI -7.36M,-1.27M]; $\Delta\text{viol.}$ -0.0145 [CI -0.0220,-0.0073]; shortage -0.637M [CI -0.976M,-0.300M]; full-MoE family -4.26M [CI -5.43M,-3.04M]	Same-model diagnostic; not cross-model optimal
Physical-bin quantile comparator	Boundary phys. 93.82M/0.1010; GWN phys. 84.31M/0.0931; gate 95.13M/0.1021	Physical bins remain competitive
Cost-ratio applicability	Active at $\rho = 5$ -10; narrows at 20; $\rho = 50$ favors global (+5.93M, +0.0035)	Moderate-cost window only
Full-sample system value	GWN/global 464.07M/0.0901; gate-bin 481.36M/0.1214	No system-wide dispatch claim
Cross-backbone/full-sample uncertainty	Δcost +17.28M; 95% CI [-52.09M,+87.95M]; $p=0.752$	Not system-wide or cross-backbone
Operational scope	Validation-frozen shortfall quantiles; no OPF, unit commitment, market clearing, or prices	Screening audit only

Same-model intervals are $n = 5$ seed-paired bootstrap mean CIs for gate-bin minus same-router global at $\rho = 10$.

Uncertainty-conditioned reserve pricing

Table A11b reports the soft-posterior reserve strategies. All strategies use validation-frozen shortfall quantiles inside the ± 1.0 m s⁻¹ boundary band; bin edges for the learned strategies are fitted on validation anchor posteriors only. P(pitch) quintiles (soft-gate-bin) carry the boundary-direction information, while gate-entropy and max-probability quintiles are pure routing-uncertainty signals that require no physical bin and no threshold. Seed-paired bootstrap CIs are reported against global and physical-bin references.

TABLE XXIV
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Table A11b. Soft-posterior reserve strategies at $\rho = 10$ (5 seeds, boundary band).

Strategy	Total cost	Violation	Reserve	Shortage
Global	16.632M	0.1000	10.948M	0.568M
Physical-bin	16.190M	0.0996	10.722M	0.547M
Soft-gate-bin (P(pitch) quintiles)	16.065M	0.1012	10.331M	0.573M
Entropy-bin (gate-entropy quintiles)	16.324M	0.1010	10.591M	0.573M
Maxprob-bin (max-probability quintiles)	16.307M	0.1014	10.530M	0.578M

Seed-paired 95% bootstrap CIs (strategy minus baseline): soft-gate-bin vs global Δcost -568k [-726k,-406k], $\Delta\text{pinball@0.9}$ -1.82 [-2.33,-1.30]; entropy-bin vs global Δcost -309k [-396k,-221k], $\Delta\text{pinball}$ -0.99 [-1.27,-0.71]; maxprob-bin vs global Δcost -325k [-454k,-214k], $\Delta\text{pinball}$ -1.04 [-1.45,-0.69]. Versus physical-bin: entropy-bin +133k [+0.8k,+256k], maxprob-bin +117k [-29k,+264k] (matches). Violation rises by about 0.001 for the learned strategies. Protocol: boundary-band anchor cells only; not interchangeable with the full-sample Table III audit.

Table A11c reports the modular-equivalence control on the same boundary-router backbone. The classifier is a validation-fit logistic regression on issue-time anchors (Wspd, Pab_mean), and its hard and soft bins are compared with the gate's. On clean anchors the classifier reproduces the physical-bin reserve almost exactly and beats the hard gate route, so the hard route has no reserve increment; the jointly-learned soft posterior remains below every modular alternative, and Table A11d localizes that pricing increment to joint learning rather than to the routing structure.

TABLE XXV
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Table A11c. Modular-equivalence reserve control on the same backbone ($\rho = 10$, 5 seeds, boundary band).

Strategy	Total cost	Δ vs global	Δ vs physical-bin
Global	16.632M	—	—
Physical-bin (rule labels)	16.190M	—	—
Classifier-bin (hard)	16.190M	-442k [-534k, -355k]	-68 [-112, -28]
Soft-clf-bin (P(pitch) quintiles)	16.234M	-398k [-483k, -324k]	+44k [+29k, +61k]
Gate-bin (hard route)	16.398M	-234k [-369k, -107k]	+208k [+93k, +332k]
Soft-gate-bin (P(pitch) quintiles)	16.065M	-568k [-726k, -406k]	-125k [-297k, +34k]

The classifier is refit per seed on validation anchors only. Hard classifier bins are nearly equivalent to physical bins on clean anchors; the hard gate route is worse than both. The soft gate posterior is the only strategy below every modular alternative (16.065M). Combined with the fair-degradation audit, where the classifier falls to recall 0.885 under the strongest noise against 0.951 for the gate, the modular classifier matches clean-anchor pricing with no detected difference while the soft posterior keeps the degraded-stream robustness in one deployment object. Script: `scripts/modular_equiv_reserve.py`.

Mechanism decomposition of the boundary-risk posterior

Table A11d consolidates the counterfactual controls that localize where the posterior's value comes from. The soft-physical pitch quantile is the honest clean-observation baseline; joint learning is what prices reserve risk below the global rule and far below an independent classifier posterior; the routing structure is what survives confirming-stream degradation. Scripts: `scripts/soft_rule_contrast.py`,

scripts/degraded_gate_reserve.py,
 scripts/kelmarsh_reserve_pricing.py,
 scripts/multichannel_classifier.py,
 scripts/gbdt_reserve_pricing.py,
 scripts/dense_classifier_pricing.py,
 scripts/fair_degradation_replay.py.

TABLE XXVI

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Table A11d. Mechanism decomposition with counterfactual controls ($\rho = 10$, 5 seeds, boundary band).

Method	Jointly trained	Needs physical channels	Router	Reserve cost	Degraded recall	Reading
Soft physical pitch quantile	no	yes	no	15.538M	–	Honest clean-observation pricing baseline; collapses toward others under degradation
Threshold rule (physical-bin)	no	yes	no	16.190M	0.196	Collapses under confirming-stream delay
Independent logistic classifier	no	yes	no	16.190M	0.885 (noise)	Equivalent clean pricing; no degradation robustness
Independent GBDT posterior	no	yes	no	17.650M	0.868	Immune to WopdPub delay but prices 1.0M worse than the global rule
Joint non-routed posterior head	yes	weak	no	16.070M	0.286	Joint learning prices; without routing it collapses under delay
Joint routed posterior (this work)	yes	weak	yes	16.855M	0.933	Joint learning prices and the routing structure survives degradation
Global quantile	–	–	–	16.632M	0.971 (clean)	Reference

Reserve costs are validation-frozen boundary-band totals (mean over five seeds); seed-paired CIs: soft-physical vs joint-routed +526k [330k, 719k]; joint-routed vs global -568k [-726k, -406k]; joint-routed vs independent GBDT -1.59M (see [artifacts/breakthrough_20260904](#)). Degraded recall is early-window recall under a six-step confirming-stream delay (or strongest noise level where noted) applied identically to every detector; the soft-physical pitch quantile is a continuous pricing rule without discrete recall, and under a 6-step delay its reserve cost degrades to 16.048M. Kelmarsh sparse-farm pricing increment is not significant (-40k, CI [-202k, +129k]) and is disclosed as such. Time-block robustness: a hierarchical seed-and-week-block bootstrap over daily reserve costs across the 35-day test period keeps the gate-bin-vs-global difference significantly negative for both MoE families (Boundary router pooled 5-seed total delta of -22.40M, corresponding to a per-seed mean of -4.48M which matches the -4.42M per-seed evaluation in Table A11 within bootstrap resampling granularity, 95% hierarchical block CI [-39.55M, -7.61M]; Physics-Aligned MoE pooled 5-seed total -23.39M, CI [-42.41M, -7.02M]), confirming that the reserve advantage is robust to temporal autocorrelation across weeks.

Kelmarsh asynchronous event-stream grounding

The confirming-stream delay scenario is grounded in the operational reality of asynchronous event streams as observed in the Kelmarsh 2016 archive. The Status stream exported by the Greenbyte platform carries 14,019 status events with second-level timestamps; 99.6% of them fall strictly between the 10-minute turbine-data periodic grid points, meaning that 10-minute periodic SCADA sampling cannot synchronously confirm operating state events at issue time. Pitch-system operating states arrive as a separate asynchronous event stream (e.g. “Pitch measuring system 1><2”). Rather than claiming a fixed runtime delay in field telemetry, we evaluate operational resilience via a controlled deployment stress test with 1- to 6-step confirmation delays and sensor noise. Script and artifact details are archived in [artifacts/kelmarsh_grounding_20260903/](#).

Unified mechanism-control statistics

Table A11e consolidates every mechanism control into one reviewer-facing statistics table with family-wise multiplicity control ($m = 13$). Both nominal 95% bootstrap CIs and Bonferroni-adjusted 99.62% CIs ($\alpha = 0.05/13$) are reported. Every control declared significant at the 95% level survives the Bonferroni adjustment without crossing zero. For the joint non-routed dense head versus the joint gate router (+0.01M, 95% CI [-0.16M, +0.19M]), no difference is detected; we do not assert formal equivalence because no equivalence margin was prespecified for TOST.

TABLE XXVII

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Table A11e. Unified mechanism-control statistics (total cost, $\rho = 10$).

Control	Δ	95% CI	Excludes zero	Condition
Soft-physical vs global	-1.09M	[-1.32M, -0.88M]	yes	WTB clean
Soft-physical vs physical-bin	-0.65M	[-0.80M, -0.50M]	yes	WTB clean
Joint gate vs global	-0.57M	[-0.73M, -0.41M]	yes	WTB clean
Joint gate vs physical-bin	-0.13M	[-0.30M, +0.03M]	no	WTB clean
Independent GBDT vs joint gate	+1.59M	[+1.40M, +1.76M]	yes	WTB clean
Joint non-routed vs joint gate	+0.01M	[-0.16M, +0.19M]	no (no detected diff.)	WTB clean
Joint gate vs global (sparse farm)	-0.04M	[-0.20M, +0.13M]	no	Kelmarsh 55%
Soft-physical vs global (sparse farm)	-0.03M	[-0.09M, +0.05M]	no	Kelmarsh 55%

$m = 13$ family; all 95%-significant controls remain significant under Bonferroni ($\alpha = 0.05/13$, 99.62% CI) because their CIs exclude zero. CI crossing zero indicates no detected difference, not formal equivalence. Raw tables and scripts in [artifacts/pl_stats_20260904/](#).

Engineering-unit reserve-value translation

Table A12 translates the reserve audit into MWh-equivalent forecast-cell accounting and an illustrative 100 EUR/MWh reserve-cost-scale marker. Its role is unit interpretation, not market valuation. Full decision links require probabilistic forecasts and system-level constraints beyond this screening protocol [3], [4].

TABLE XXVIII

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Table A12. Engineering-unit reserve-value translation at $\rho = 10$.

Comparison	Δ reserve	Avoided shortage	Δ cost	Cost-scale@100
Boundary gate-bin vs same-router global (train-only)	+1950	+637	-4420	-442k
Boundary gate-bin vs GWN physical-bin (legacy)	+2319.2	+205.6	+262.9	+26k
Boundary gate-bin vs GWN global full sample (legacy)	+3148.2	-1413.5	+17283.2	+1728k

Values are MWh-equivalent forecast-cell accounting with $\Delta t = 1/6$ h.

Cost-scale@100 is an illustrative reserve-cost scale marker at 100 EUR/MWh, not market revenue, settlement value, OPF, or unit-commitment output. Boundary rows are from the train-only rerun; GWN comparison rows remain legacy pure-prediction baselines. Use same-router/global as the bounded boundary-window diagnostic; physical-bin and full-sample rows block reserve superiority.

Complete forecasting benchmark

Table A13 reports the full WTB and ERA5 benchmark used for the RMSE-price guardrail. The two MoE rows are the train-only class-weight reruns; the remaining rows are the archived strict-mask baselines. The displayed guardrail is the 5.59 RMSE gap between the train-only boundary router and iTransformer.

TABLE XXIX
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Table A13. Complete forecasting benchmark (WTB RMSE in kW; ERA5 normalised flux RMSE).

Model	Overall RMSE	Switch RMSE
iTransformer	224.34 $\pm$ 2.23	228.80 $\pm$ 2.54
Graph WaveNet	225.74 $\pm$ 2.60	228.12 $\pm$ 3.23
TiDE	227.31 $\pm$ 3.05	231.48 $\pm$ 2.86
PatchTST	228.07 $\pm$ 4.05	231.22 $\pm$ 4.15
Boundary-forced router (train-only)	229.93 $\pm$ 2.50	231.65 (5 seeds)
Capacity-Matched Dense	231.42 $\pm$ 5.87	233.82 $\pm$ 5.96
Physics-Aligned MoE (train-only)	231.90 (5 seeds, 225.35-236.21)	–
Graph Transformer	235.38 $\pm$ 5.15	237.06 $\pm$ 5.78
Unconstrained MoE	235.40 $\pm$ 7.96	237.22 $\pm$ 8.11
GAT-GRU	236.59 $\pm$ 7.80	240.17 $\pm$ 8.50
Boundary-forced router (legacy)	236.13 $\pm$ 8.41	239.86 $\pm$ 8.85
Physics-Aligned MoE (legacy)	241.42 $\pm$ 4.28	242.60 $\pm$ 6.07
ERA5 Graph WaveNet	0.8417 $\pm$ 0.0661	0.7934 $\pm$ 0.0652
ERA5 Physics-Aligned MoE	0.8735 $\pm$ 0.0536	0.8507 $\pm$ 0.0363
ERA5 TCN	0.8784 $\pm$ 0.0351	0.8338 $\pm$ 0.0292
ERA5 PatchTST	0.8817 $\pm$ 0.0827	0.8248 $\pm$ 0.0842
ERA5 STGCN	0.9893 $\pm$ 0.1646	0.8908 $\pm$ 0.1239
ERA5 Persistence	0.7127	0.6814

The two train-only MoE rows are the provenance-corrected reruns used for the headline guardrail; their MAE and switch statistics are available in the archived run tables. The legacy MoE rows remain listed for provenance transparency only.

ERA5 observability contrast

The ERA5 contrast tests a setting in which the physical marker is directly visible. Three archived months of hourly data over a fixed 16×16 patch are reorganised into a $T=2208$ -frame tensor with $N=256$ nodes and eight input variables. The thermodynamic regime label separates stable from convective near-surface states using sensible heat flux and its local time delta: convective samples have positive afternoon flux anomaly, stable samples do not; the label is available at 60% coverage and enters only alignment supervision. The gate anchor is `[sshf, t2m, wind_speed,  $\Delta$ sshf]`, and the graph is a static Haversine-Gaussian k_{nn} structure because no wake directionality is defined on a grid. The same directed-diffusion GRU and node-level MoE are trained with the full corrected stack; the contrast result is that Physics-Aligned MoE achieves NMI 0.210 ± 0.109 against an unconstrained collapse of 0.032 ± 0.025 under thermodynamic observability (`table_routing_quality.csv`), showing that supervision retains partial alignment under direct thermodynamic observation whereas unconstrained gating collapses. The ERA5 rows in Table A13 and the shared-architecture table carry the numerical record; ERA5 is not used for any reserve or deployment claim.

Compute disclosure

All neural runs use a single GPU (NVIDIA RTX 4090), mixed precision, AdamW, gradient clipping, and early stopping on validation RMSE. The WTB boundary router has 110,012 parameters; one epoch takes approximately 445 s and early stopping selects checkpoints at 6-9 epochs, so a five-seed family costs roughly 6-7 GPU-hours. The signature-gate probes, fair-degradation replays, and reserve audits reuse saved checkpoints and test arrays and are CPU-minute analyses except the replay forwards, which re-evaluate the test split

in under ten minutes on one GPU. All experiments are reproducible across the declared seeds; bitwise reproducibility across CUDA environments is not claimed.

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